Sanskar

Full-Stack Developer | Web Applications & Systems

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LinkedIn | GitHub | LeetCode

Education

Indian Institute of Information Technology (IIIT), Dharwad — Dharwad, Karnataka

B.Tech in Electronics and Communication Engineering | Minor: Cyber Security | CGPA: ***8.02/10*** — *2023 – 2027*

Relevant Coursework: Data Structures and Algorithms, Object-Oriented Programming, Database Management Systems, Operating Systems, Computer Networks

Technical Skills

Languages: JavaScript, Python,TypeScript, C++, C
Frontend: React, Next.js, HTML5, CSS3,Tailwind CSS
Backend: Node.js, Express.js, REST APIs, FastAPI, Microservices
Databases: MySQL, PostgreSQL, MongoDB, Redis, Prisma
Developer Tools & Cloud: Git, GitHub, Docker, Kubernetes, CI/CD, Vercel, RabbitMQ, Railway
Core Concepts: Operating Systems, Microservice Architecture, Data Structures and Algorithms, OOP, DBMS, Computer Networks, Distributed Systems, System Design
Problem Solving: 300+ problems solved on LeetCode (Arrays, Trees, Graphs, DP, Binary Search, Sliding Window)

Technical Experience

WorldQuant BRAIN — Quantitative Research Consultant (Gold Level) — Remote

10-2025 – 06-2026

- Developed and back-tested **100+ quantitative strategies** using Python, time-series analysis and historical market data.
- Built reusable workflows to evaluate signal quality using Sharpe ratio, turnover, and drawdown metrics.
- Earned **Gold Level** on the WorldQuant BRAIN platform through consistent strategy development and submissions.

Projects

CorePay — Fintech Payment Platform — 2026

Node.js, TypeScript, Express, PostgreSQL, Prisma, Redis, RabbitMQ, Docker

- Built a **microservices-based payment platform** with Node.js/TypeScript, API Gateway, JWT/RBAC authentication, rate limiting, and circuit-breaker based resilience.
- Implemented **reliable payment processing** using idempotency keys, Redis distributed locking, PostgreSQL transactions, refunds, and a double-entry ledger with integer-based monetary units.
- Designed an **event-driven architecture** using RabbitMQ, asynchronous notifications, retries, dead-letter queues, HMAC-secured webhooks, and Docker-based deployment with observability and load testing.

VeriSync — Attendance Management System — 2026

TypeScript, React, Vite, NestJS, Prisma, PostgreSQL, Tailwind CSS

- Successfully **delivered and deployed an early-stage production system** to a college, providing a full-stack attendance platform with role-based portals for admins, teachers, and students.
- Built a **multi-layered attendance verification workflow** using dynamic QR codes, OTPs, and facial recognition to prevent proxy attendance across 25+ subjects.
- Developed a **real-time analytics dashboard** with React and Recharts to monitor attendance trends, generate insights, and manage correction requests.
- Optimized backend queries and database operations to support **concurrent check-ins for 150+ students** with sub-200ms API response times.
- Collaborated with college stakeholders during the **initial deployment and validation phase**, incorporating real-world feedback to improve usability, reliability, and attendance workflows.

CareerOS — AI-Powered Career Intelligence Platform — 2026

FastAPI, Next.js, PostgreSQL, pgvector, Redis, Celery, OpenAI API

- Built a **production-oriented career platform** combining resume building, ATS analysis, semantic job matching, and RAG-powered career assistance using FastAPI, Next.js, and PostgreSQL.
- Designed a **canonical versioned resume architecture** with PostgreSQL, immutable JSONB snapshots, ownership-enforced APIs, autosave, section reordering, and multi-template PDF/DOCX export.
- Implemented a **hybrid RAG retrieval pipeline** using PostgreSQL full-text search, pgvector embeddings, Reciprocal Rank Fusion (RRF), and cross-encoder reranking for context-aware career recommendations.
- Built an **asynchronous AI processing architecture** with Redis and Celery for document ingestion, resume analysis, embeddings, ATS evaluation, and background generation workloads.

Achievements

- **Wall Street Algo Trading Challenge 2025 — 1st Place:** Built the top-ranked quantitative strategy in a challenge organized by The Wall Street Club, BITS Pilani Hyderabad, and AlgoAnalytics.
- **YUKTI Chip Design Challenge 2025 — 1st Place:** Implemented a complete ASIC design flow from RTL to GDSII using Verilog.
- **Vidyut Vega Hackathon 2025 — 3rd Place:** Optimized a deep learning model for deployment on resource-constrained edge hardware.